

Leucine signaling in the pathogenesis of type 2 diabetes and obesity

Epidemiological evidence points to increased dairy and meat consumption, staples of the Western diet, as major risk factors for the development of type 2 diabetes (T2D). This paper presents a new concept and comprehensive review of leucine-mediated cell signaling explaining the pathogenesis of T2D and obesity by leucine-induced over-stimulation of mammalian target of rapamycin complex 1 (mTORC1). mTORC1, a pivotal nutrient-sensitive kinase, promotes growth and cell proliferation in response to glucose, energy, growth factors and amino acids. Dairy proteins and meat stimulate insulin/insulin-like growth factor 1 signaling and provide high amounts of leucine, a primary and independent stimulator for mTORC1 activation. The downstream target of mTORC1, the kinase S6K1, induces insulin resistance by phosphorylation of insulin receptor substrate-1, thereby increasing the metabolic burden of β -cells. Moreover, leucine-mediated mTORC1-S6K1-signaling plays an important role in adipogenesis, thus increasing the risk of obesity-mediated insulin resistance. High consumption of leucine-rich proteins explains exaggerated mTORC1-dependent insulin secretion, increased β -cell growth and β -cell proliferation promoting an early onset of replicative β -cell senescence with subsequent β -cell apoptosis. Disturbances of β -cell mass regulation with increased β -cell proliferation and apoptosis as well as insulin resistance are hallmarks of T2D, which are all associated with hyperactivation of mTORC1. In contrast, the anti-diabetic drug metformin antagonizes leucine-mediated mTORC1 signaling. Plant-derived polyphenols and flavonoids are identified as natural inhibitors of mTORC1 and exert anti-diabetic and anti-obesity effects. Furthermore, bariatric surgery in obesity reduces increased plasma levels of leucine and other branched-chain amino acids. Attenuation of leucine-mediated mTORC1 signaling by defining appropriate upper limits of the daily intake of leucine-rich animal and dairy proteins may offer a great chance for the prevention of T2D and obesity, as well as other epidemic diseases of civilization with increased mTORC1 signaling, especially cancer and neurodegenerative diseases, which are frequently associated with T2D.